

Wazuh SIEM Home Lab

Beginner Blue-Team / SOC Analyst Portfolio Report

This report documents the setup, troubleshooting, and successful access of a Wazuh SIEM lab built on Ubuntu Server in Oracle VirtualBox.

Project Type	Home SIEM / Blue-Team Lab
SIEM Platform	Wazuh 4.14.x
Server OS	Ubuntu Server 24.04 LTS
Host System	Windows 11 with Oracle VirtualBox
Final Status	Dashboard installed, reachable, and login verified

1. Executive Summary

The lab successfully deployed Wazuh as a SIEM platform inside a VirtualBox Ubuntu Server VM. The final dashboard was accessed from the Windows host through localhost port forwarding, and the Wazuh overview page loaded after successful authentication.

The most valuable part of the project was not only installation, but troubleshooting: unsupported OS selection, disk allocation problems caused by LVM defaults, VM resource tuning, and browser access through VirtualBox networking.

2. Lab Objectives

- Build a working local SIEM lab suitable for entry-level SOC and blue-team practice.
- Install Wazuh on Ubuntu Server 24.04 LTS inside Oracle VirtualBox.
- Learn how VM CPU, memory, and disk layout affect security tooling.
- Access the Wazuh dashboard securely from the Windows host browser.
- Prepare the environment for future Windows endpoint monitoring with Wazuh Agent and Sysmon.

3. Environment

Layer	Configuration
Host	Windows 11
Hypervisor	Oracle VirtualBox
SIEM VM	Ubuntu Server 24.04 LTS
SIEM Stack	Wazuh Manager, Wazuh Indexer, Filebeat, Wazuh Dashboard
VM Resources	Approximately 6 GB RAM, 3 CPU cores, larger disk layout with about 68 GB free
Dashboard URL	https://127.0.0.1:8443 via VirtualBox port forwarding

4. Implementation Timeline

- Created an Ubuntu Server VM in VirtualBox and configured CPU, memory, and disk.
- Installed Ubuntu Server and confirmed network connectivity using IP and ping checks.
- Downloaded and ran the Wazuh installation assistant.
- Investigated repeated dashboard failures and identified two root causes: unsupported Ubuntu version and insufficient root partition size.
- Rebuilt the lab with Ubuntu Server 24.04 LTS and a larger usable root filesystem.
- Installed Wazuh successfully and captured the generated administrator credentials privately.
- Configured browser access to the dashboard through VirtualBox NAT port forwarding.
- Logged in and confirmed the Wazuh overview page loaded.

5. Troubleshooting Record

Issue: Ubuntu Version Mismatch

The first installation attempt used Ubuntu 26.04, which produced Wazuh compatibility warnings. The lab was rebuilt using Ubuntu Server 24.04 LTS to align with the supported operating system path.

Issue: Root Partition Too Small

The initial LVM-based storage layout allocated only about 24 GB to the root filesystem. Wazuh Dashboard is storage-heavy during installation, so this caused repeated failures. The fix was to rebuild with a larger virtual disk and verify roughly 68 GB of free space before reinstalling.

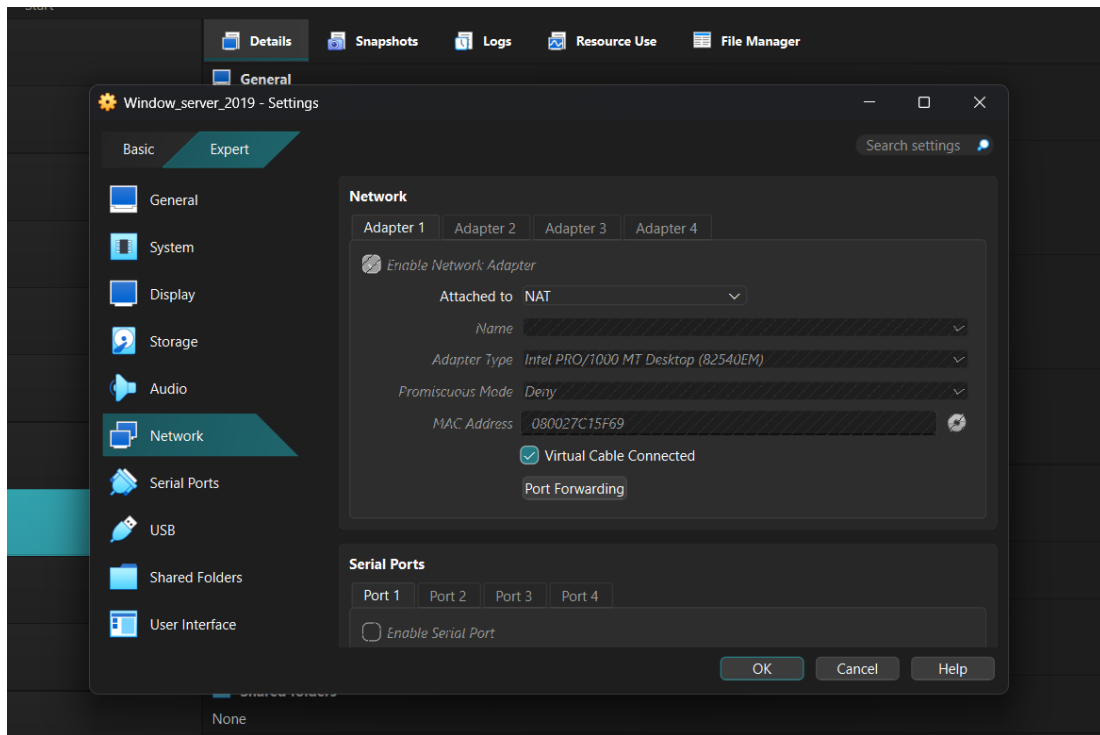
Issue: Browser Access from Windows Host

The Wazuh VM used VirtualBox NAT networking, so the dashboard was exposed to the Windows host using port forwarding. The dashboard was then opened at <https://127.0.0.1:8443> in the Windows browser.

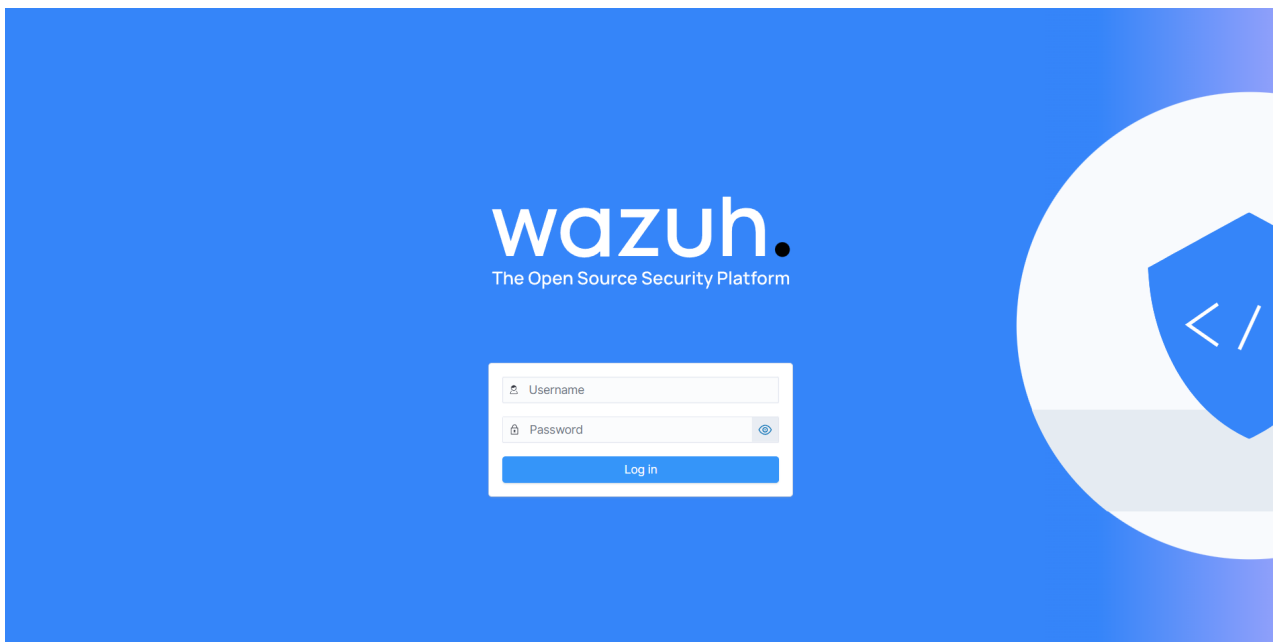
6. Evidence Screenshots



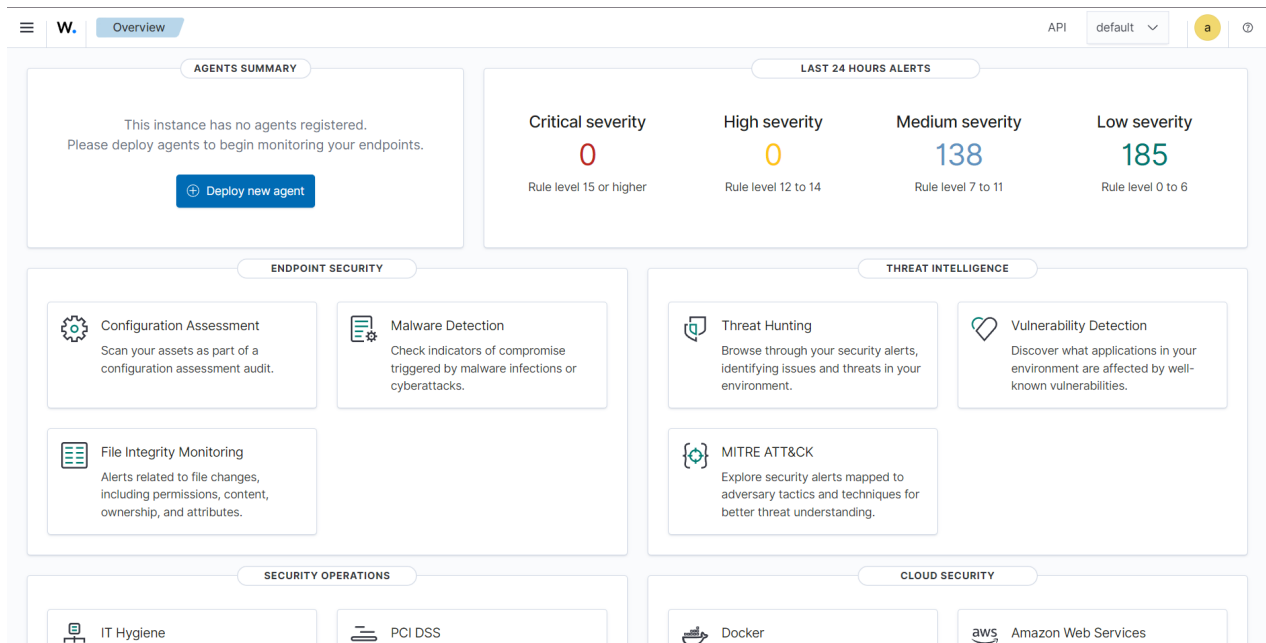
Wazuh dashboard loading screen



VirtualBox Windows Server network review



Wazuh login page reached through localhost port forwarding



Wazuh overview dashboard after successful login

7. Security and Privacy Notes

- Wazuh administrator credentials are not included in this report.
- Password files and secrets should not be committed to GitHub or shared publicly.
- Screenshots should be reviewed before posting to ensure no credentials or sensitive host data are visible.
- The lab is for defensive learning and should be expanded using owned VMs or explicitly authorized systems only.

8. Skills Demonstrated

- Linux server installation and command-line operation.
- VirtualBox VM provisioning and resource tuning.
- SIEM deployment with Wazuh.
- Storage troubleshooting and root filesystem verification.
- NAT networking and local port forwarding.
- Security documentation suitable for a beginner SOC portfolio.

9. Next Steps

- Add a Windows Server or Windows 11 VM as a Wazuh agent.
- Install Sysmon and collect process creation, network, and PowerShell telemetry.
- Generate safe test events such as failed logins and suspicious command execution.
- Write short incident reports using Wazuh alerts as evidence.
- Publish the GitHub README and attach this PDF as project documentation.

Conclusion: The SIEM foundation is operational. The next portfolio milestone is endpoint telemetry and detection writeups.